of the Cosmos (Barrow 2011); and Chapters 3, 14, and 16 of *From Eternity to Here: The Quest for the Ultimate Theory of Time* (Carroll 2011). For current research on the big bang, see the blog by Sean Carroll, *Preposterous Universe* (Carroll 2014) at http://www.preposterousuniverse.com/blog/.

Chapter 3. The Laws That Control the Universe

Richard Feynman, one of the great physicists of the twentieth century, gave a series of lectures for the general public in 1964 that delved deeply into the nature of the laws that control our universe. He wrote up his lectures in one of my favorite books of all time, *The Character of Physical Law* (Feynman 1965). For a more detailed, more up-to-date, and much longer book on the same topic, see *The Fabric of the Cosmos: Space, Time, and the Texture of Reality* (Greene 2004). Easier going, perhaps more fun, and equally deep is *The Grand Design* (Hawking and Mlodinow 2010).

Chapter 4. Warped Time and Space, and Tidal Gravity

For historical details on Einstein's concepts of warped time and space, their connection to tidal gravity, and his relativistic laws built on these concepts, see Chapters 1 and 2 of *Black Holes & Time Warps: Einstein's Outrageous Legacy* (Thorne 1994); and for a plethora of experiments that show Einstein was right, see *Was Einstein Right? Putting General Relativity to the Test* (Will 1993). *"Subtle Is the Lord . . .": The Science and the Life of Albert Einstein* (Pais 1982) is a biography of Einstein that focuses in depth on all of Einstein's contributions to science; it's much tougher going and much more scholarly than Thorne or Will. There are other, more comprehensive biographies of Einstein—I especially like *Einstein: His Life and*